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SURVEY

Deep Learning in Automatic Diabetic Retinopathy Detection and Grading Systems: A Comprehensive Survey and Comparison of Methods

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ABSTRACT Diabetic Retinopathy is one of the leading global causes of vision impairment and blindness in humans. It has seen a rise in prevalence, necessitating the development of advanced automatic detection methods. This paper presents a survey of the evolution in deep learning techniques for diabetic retinopathy detection, emphasizing the transition from traditional machine learning to sophisticated deep learning architectures such as convolutional neural networks. It discusses the role of transfer learning, end-to-end learning, and hybrid models in overcoming medical detection challenges while highlighting the need for artificial intelligence interpretability and real-time screening integration in clinical workflows. Building on this survey, the paper introduces a focused study on cross-dataset deployment of transfer learning for diabetic retinopathy detection and grading. Consequently, this paper evaluates 26 pre-trained models from various convolutional neural network families to provide a comprehensive comparison between the state-of-the-art CNN architectures in the field. Additionally, this study also employs Grad-CAM visualization to interpret the model's decision-making, bridging advanced artificial intelligence techniques with practical healthcare applications for diabetic retinopathy.

INDEX TERMS Diabetic retinopathy detection, deep learning, convolutional neural networks, transfer learning.

I. INTRODUCTION

Diabetic retinopathy is a diabetes-induced eye disease and is recognized as one of the leading causes of blindness among the working-age population worldwide [1]. This condition, characterized by damage to the retinal blood vessels, progresses silently and often remains undetected until vision impairment becomes severe [2]. In this condition, blood vessels in the retina inflate, leak, or clot giving rise to the formation of new blood vessels [3]. Diabetic retinopathy (DR) can be detected by the evaluation of injured blood vessels and the formation of new ones behind the retina through fundus imaging [4]. Globally, approximately 27% of diabetes patients suffer from diabetic retinopathy [5], and its global prevalence underscores the critical need for effective and early detection methods. Early intervention is vital as it significantly reduces the risk of severe vision loss. Recent studies have shown that over 90% of the cases of diabetes-induced blindness can be prevented if DR is diagnosed in its early stage [6]. However, the traditional methods of DR screening, which rely on the

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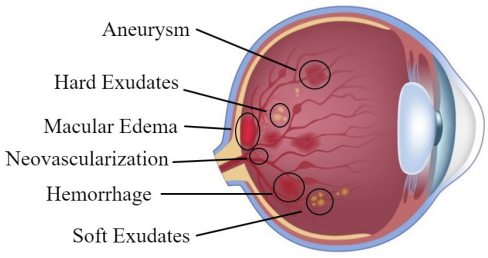


FIGURE 1. Lesions found in patients with diabetic retinopathy.

manual examination of fundus images by ophthalmologists, are time-consuming, subjective, error-prone, and often limited by the availability of expert resources [7].

Diabetic retinopathy affects the light-sensitive tissue at the back of the retina [4]. It is caused by the damage to the blood vessels of the retina due to prolonged high blood sugar levels [8]. As one of the most common causes of vision impairment and blindness in individuals with diabetes, DR presents a significant public health challenge. The progression of DR is often asymptomatic in its early stages and is presented with minimal symptoms [9], making early detection and timely intervention crucial to prevent irreversible vision loss. Additionally, DR is a prime candidate for other eye-related complications such as Macular Edema which implies swelling or thickening of the macula. This can lead to blurry vision and occur at any stage of DR. Detection and grading of DR is therefore paramount in the medical field. The identified and extracted features from retinal images are based on specific structures or abnormalities in the retinal image that are indicative of DR as presented in Figure 1. The main features of interest in DR detection include Microaneurysms that is presented by small red dots in the retinal image; Hemorrhages which are red patches that are larger than microaneurysms; Exudates which are yellowish, lipid-residue spots; and Neovascularization which refers to the growth of new, abnormal blood vessels. This disease progresses through several stages as presented in Figure 2, each marked by increased severity and risk of vision impairment [10]. The main stages of DR are Mild, Moderate, Severe Nonproliferative Retinopathy (NPDR), and Proliferative Diabetic Retinopathy (PDR) [11]. The details of each stage are presented in Table 1.

A. IMPORTANCE OF AUTOMATIC EARLY DETECTION OF DIABETIC RETINOPATHY

Early detection of DR allows for timely medical interventions, such as laser treatment, anti-VEGF injections, or vitrectomy, which can significantly reduce the risk of severe vision loss [12]. Traditional screening methods, relying on manual examination by ophthalmologists, are not only labor-intensive but also constrained by the availability of skilled professionals [12]. Furthermore, the subjective nature of these examinations can lead to variability in diagnoses. The integration of machine learning and deep learning techniques in DR screening represents a paradigm shift, offering the potential for automated, accurate, and efficient diagnostic processes. These technologies can enhance the screening process by providing consistent and objective assessments, thereby improving the clinical management of diabetic patients and reducing the burden of DR-related vision impairment on a global scale [9].

B. CURRENT CHALLENGES AND RESEARCH GAPS

Given the complexity of the disease and the intricacies of neural network methodologies, this section presents some of the primary challenges faced by researchers in the field of deep learning. Figure 3 presents a summary of the challenges and research gaps in the field of DR and their possible solutions.

1) CLASS IMBALANCE

Many datasets used for training DR detection models suffer from class imbalance, where normal images significantly outnumber those with DR. This imbalance can lead to models that are biased towards predicting the majority class, reducing their effectiveness in identifying actual cases of DR. Future research should focus on developing robust methods to handle class imbalance in training datasets. Techniques like synthetic data generation or advanced sampling methods could provide more balanced datasets for training, leading to more accurate and fair models.

2) DATASET VARIABILITY AND GENERALIZABILITY

Variability in datasets, arising from differences in imaging equipment, patient demographics, and image acquisition protocols, poses a challenge. This variability can affect the generalizability of the models, limiting their performance across diverse real-world settings. Many models perform exceptionally well on the datasets they are trained on but struggle to maintain this performance on new and unseen datasets. Furthermore, given the complexity of neural networks, they can be easily overfitted to training data, reducing their generalization ability to unseen data. Ensuring that models can generalize effectively across different populations and imaging conditions is crucial for their clinical applicability. Enhancing the ability of models to generalize across different datasets is a critical area of focus. Research into domain adaptation and transfer learning strategies could improve the robustness of models when applied to diverse populations and imaging conditions.

3) DATA AVAILABILITY AND QUALITY

High-quality, annotated fundus images for training neural networks are limited in availability. Data collection is needed to produce diverse datasets and cross-examination by expert ophthalmologists is needed to annotate the data.

4) INTERPRETABILITY

Neural networks, particularly deep learning models, often act as ‘‘black boxes,’’ making it challenging to interpret their decision-making processes. Clinicians might be hesitant to



FIGURE 2. Stages of diabetic retinopathy.

TABLE 1. Description and lesions associated with each stage of diabetic retinopathy.

Severity	Description	Lesions Present
Mild NPDR	The initial stage, characterized by the presence of small areas of balloon-like swelling in the retina's tiny blood vessels. Retinal blood vessels weakened.	Only microaneurysm.
Moderate NPDR	The second stage, where some blood vessels that nourish the retina become blocked. The retina may begin to show signs of distorted structure and function.	Intra-retinal hemorrhage, venus beading, or exudates in at least one quadrant.
Severe NPDR	The third stage, where more blood vessels are blocked. Several areas of the retina are deprived of blood supply. Advent of growth of new blood vessels.	More than 20 hemorrhages, venus beading in more than one quadrant, intra-retinal microvascular abnormalities.
PDR	This advanced stage is characterized by the growth of several new tiny blood vessels on the retina. These new vessels are fragile and can leak blood, leading to serious vision problems, including blindness.	Neovascularization and hemorrhages in the vitreous and preretinal regions.

trust a model's diagnosis without understanding its reasoning. There is a growing need for interpretable AI models in healthcare. Enhancing the interpretability of machine learning (ML) and deep learning (DL) models in DR detection will increase trust among clinicians and patients and provide insights into the decision-making process of these models.

5) INTEGRATION INTO CLINICAL WORKFLOW

In clinical settings, models must offer rapid predictions, which can be challenging for complex neural networks. Integrating neural network solutions with existing medical systems or databases requires seamless interfacing. Developing real-time DR screening systems would greatly benefit primary care settings and remote areas. Such systems could provide instant feedback and early warning signs, leading to timely referrals and treatments.

6) AUTOMATED DISEASE PROGRESSION PREDICTION

Another avenue of research is the development of automated systems for disease progression prediction. Such systems could assist in monitoring DR progression and tailoring treatment plans more effectively.

C. OBJECTIVES AND PAPER ORGANIZATION

The advent of machine learning and deep learning technologies offers a transformative solution to these challenges, providing automated, accurate, and efficient tools for DR detection and classification. The objective and motivation of this survey is to delve into and synthesize the recent advancements in DL methodologies applied to DR detection and grading. In the last few years, these fields have seen remarkable growth, leading to innovative approaches that promise to enhance the accuracy, efficiency, and accessibility of DR screening. By reviewing various algorithms, models, and techniques employed in recent studies, this survey aims to provide a comprehensive understanding of the current state of artificial intelligence-driven DR diagnostics. Furthermore, this work also seeks to identify gaps in the existing literature and propose directions for future research.

It is acknowledged in this paper that there have been a few recent reviews and surveys of diabetic retinopathy detection using computer-aided diagnosis such as the ones presented in articles [3], [4], [10], and [12]. Fundamentally, this work differs from those reviews and surveys regarding its organization, comprehensiveness, and comparison criteria. Similar to the works presented in [4] and [10], this paper delves into diabetic retinopathy detection and grading. However, unlike the works in [4] and [10], a comprehensive analysis of the challenges in the field of DR grading along with some possible solutions are included. Furthermore, this work aims to exclusively deep-dive into DL architectures for diabetic retinopathy detection and grading that have been developed very recently. Therefore, unlike, the reviews presented in [3] and [12], traditional non-machine learning and machine

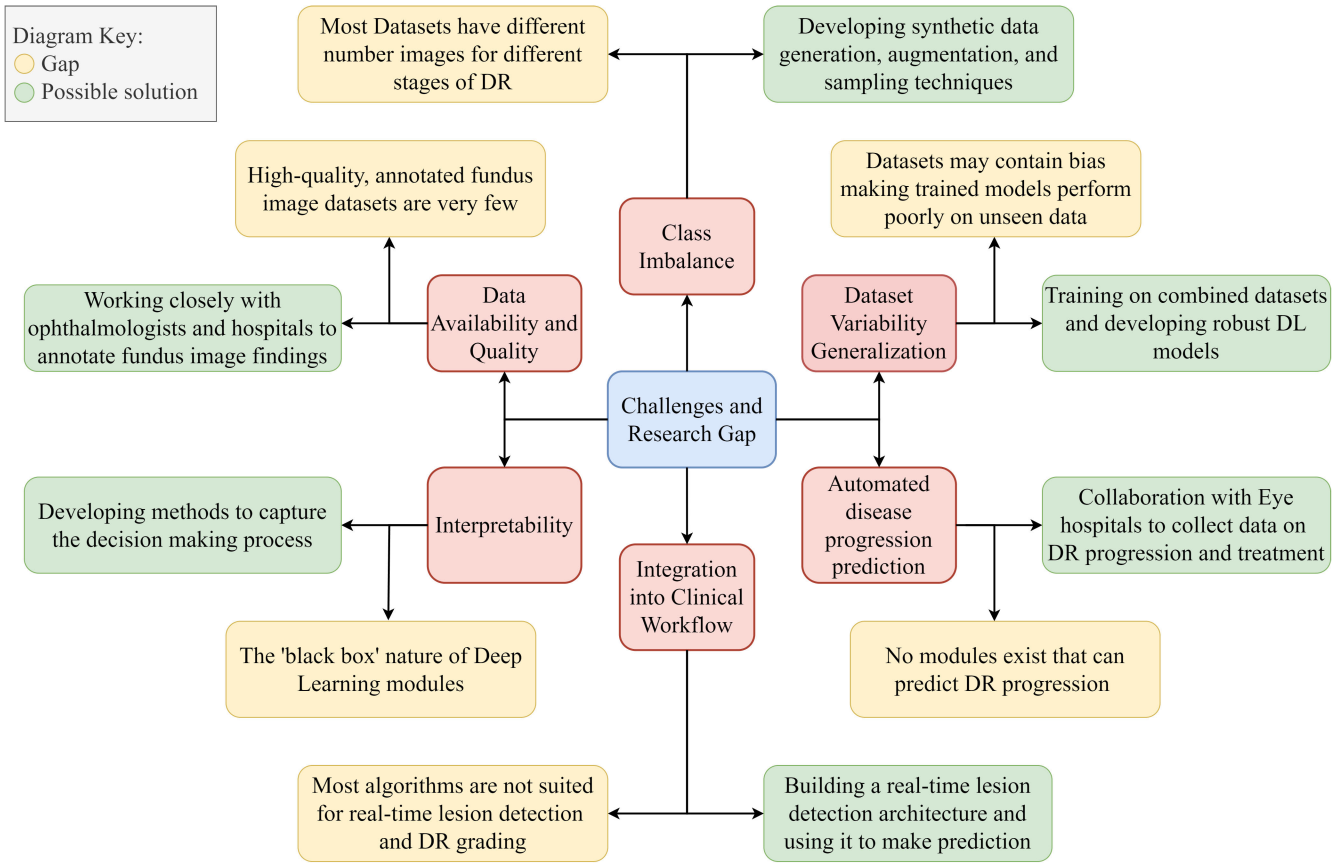


FIGURE 3. Challenges and possible solutions in the field of DR diagnosis.

learning approaches are excluded from this review. This endeavor is facilitated to concentrate more on state-of-the-art methods that involve convolutional neural networks (CNN), hybrid architectures such as transformers and ensemble learning, and the amalgamation of DL and unsupervised learning methods into one architecture. The paper also harbors vivid details on publicly available datasets and data preprocessing, which further adds to its comprehensiveness.

Thus, this paper is organized with Section II discussing various publicly available datasets for DR diagnosis, lesion detection, and DR grading. Section III provides an overview of the early machine and deep learning approaches followed by a comprehensive survey of the state-of-the-art architectures for DR detection and grading. Section IV describes the methodology for the comparison of several deep learning architectures in DR detection and classification. Section V provides an overview of the results from the experimentation on two distinct datasets. Finally, Section VI concludes the work, summarizing the key findings and emphasizing the significance of DL in advancing DR detection and classification.

II. AVAILABLE DATASETS FOR DIABETIC RETINOPATHY DETECTION AND GRADING

In the area of DR detection and classification research, the choice of datasets plays a crucial role in developing, testing, and validating deep learning models. The datasets provide the necessary input for training and testing models. Some publicly available datasets are surveyed in this section. These datasets have become the gold standard in DR research due to their size, diversity, and the complexity of the cases they include. For our survey, we have only considered datasets with at least 500 fundus images. Some of the most commonly used datasets are:

A. APTOS 2019 BLINDNESS DETECTION

The APTOS dataset [13], from the Asia Pacific Tele-Ophthalmology Society (APTOS), includes high-quality retinal images and is used for the detection and grading of diabetic retinopathy. The dataset has been collected by the Aravind Eye Hospital in India. It features a diverse set of images, which are crucial for training models to handle various lighting conditions, camera models, and ethnicities. The dataset consists of 3,662 images divided into five classes of which there are 1,805 fundus images for healthy eyes, 370 images for mild NPDR, 999 images for moderate NPDR, 193 images for severe NPDR, and 295 images for PDR.

B. MESSIDOR

The Messidor dataset [14] comprises digital color fundus photographs and includes annotations for the presence and severity of DR. It is widely used for benchmarking algorithms due to its detailed annotations and high-quality images. This dataset originally contained 1,200 color fundus images

annotated by three retina specialists. A later iteration improved the dataset by including more images of different types of DR and boosted the data to 1,748 graded images. This updated dataset is called Messidor-2.

C. EyePACS

The EyePACS dataset [15] is part of the Kaggle Diabetic Retinopathy Detection competition. This dataset includes a large number of retinal images sourced from the EyePACS screening program. It is notable for its extensive collection of images, which allows for robust training and testing of models. These datasets have been instrumental in advancing DR detection research, providing researchers with a wealth of data to develop and refine algorithms. The dataset consists of 88,702 color fundus images and is therefore the largest DR grading dataset available publicly to date. Each fundus image in this dataset has been annotated by experts in the field and contains the severity rating of diabetic retinopathy, ranging from 0 to 4. Here, 0 represents healthy eyes, 1 represents mild NPDR, 2 represents moderate NPDR, 3 represents severe NPDR, and 4 represents PDR.

D. INDIAN DIABETIC RETINOPATHY IMAGE DATASET

The Indian diabetic retinopathy image dataset (IDRiD) [16] contains 516 color fundus images from an eye clinic in Nanded, India. The images are of a high resolution of 4288 × 2848 pixels and can be used for both DR grading and lesion detection. The distribution of the dataset is such that there are 167 normal eye images, 25 images for mild NPDR, 167 images for mild NPDR, 94 images for severe NPDR, and 63 images for PDR. The dataset is highly imbalanced with very little information on mild NPDR and PDR.

E. DATASET FOR DIABETIC RETINOPATHY

The Dataset for Diabetic Retinopathy (DDR) [17] has been collected from 147 hospitals in China and is the second-largest publicly available dataset for DR grading and lesion detection. The dataset originally contained 13,673 fundus images with varying qualities and resolution. However, in later years, 1,121 images have been removed which were of poor quality to make the total number of fundus images 12,552.

III. DEEP LEARNING APPROACHES FOR DIABETIC RETINOPATHY DIAGNOSIS

Machine Learning (ML) has long been a cornerstone in the development of automated diagnostic tools for Diabetic Retinopathy (DR) [18]. These approaches, particularly in the early stages, leveraged traditional algorithms and feature extraction techniques to analyze retinal images. The first ventures into using machine learning for diabetic retinopathy detection employed algorithms such as decision trees, support vector machines (SVMs), and k-nearest neighbors (KNN) [19]. These algorithms were primarily used to classify retinal images to distinguish between normal and diseased retinas [20]. The work in [18] developed a machine-learning framework to detect and grade DR, using a dataset of 1361 fundus camera images. By applying SVM and K-NN algorithms for exudates, an accuracy of 92.1%, and a remarkable 99.9% accuracy with a decision tree algorithm for microaneurysms were achieved. These algorithms, though showing good performance in detecting lesions or the binary classification of DR, are not essentially used in DR grading due to their limitation in learning capabilities.

As the field progressed, the advent of DL opened new avenues for DR detection, addressing many of these limitations. The transition from ML to DL in DR detection was primarily fuelled by the increasing availability of large, annotated datasets and advancements in computational power. The emergence of DL in the field of medical imaging, particularly for DR detection was driven by the inherent ability of DL models, especially CNNs, to automatically learn complex and hierarchical features from raw abundant image data. This led to more accurate and robust diagnostic systems compared to traditional ML classifiers. Convolutional Neural Networks, began to outperform traditional ML approaches by learning features directly from the data, reducing the need for manual feature extraction and improving the ability to generalize across diverse datasets. These CNN-driven DL models excelled in automatically extracting and learning features from raw images, thus eliminating the need for manual feature engineering, and enabling more nuanced and detailed image analysis. These models can capture diverse intrinsic features on a global image level, resulting in the ability to grade DR and classify fundus into severity categories which was challenging for conventional ML methods. Although methods such as the ones presented in [21] and [22], were good in grading DR without using CNN, the number of steps needed before using dense layers for classification is immense. The reason is that manual features have to be extracted that rely on the geometry and statistics and this requires significant noise cancellation and segmentation techniques to transform the image. While most DL architectures have a few preprocessing steps to enhance the image, heavy noise removal techniques are not usually needed for gold-standard datasets. A summary of the evolution of machine and deep learning approaches in diabetic retinopathy diagnosis is shown in Figure 4.

Architectures that use CNN encoders have become the backbone of image analysis in DR detection due to their ability to capture spatial hierarchies in images. Advancements in the field of DR detection and diagnosis have been marked by significant contributions from various researchers utilizing deep learning techniques. There have been a few notable reviews and surveys such as the ones presented in [4], [7], [23], [24], [25], and [26] in the last few years on several DL methods that can effectively detect and grade DR in fundus imaging. These studies not only demonstrate the potential of artificial intelligence in medical imaging but also pave the way for more efficient and accurate diagnosis methods. However, in light of the fast-changing designs for diagnosis architectures, several upgraded works have come forward in the last few years. In our survey, we highlight a few CNN-based methods that have yielded improvements in the automatic classification for DR,

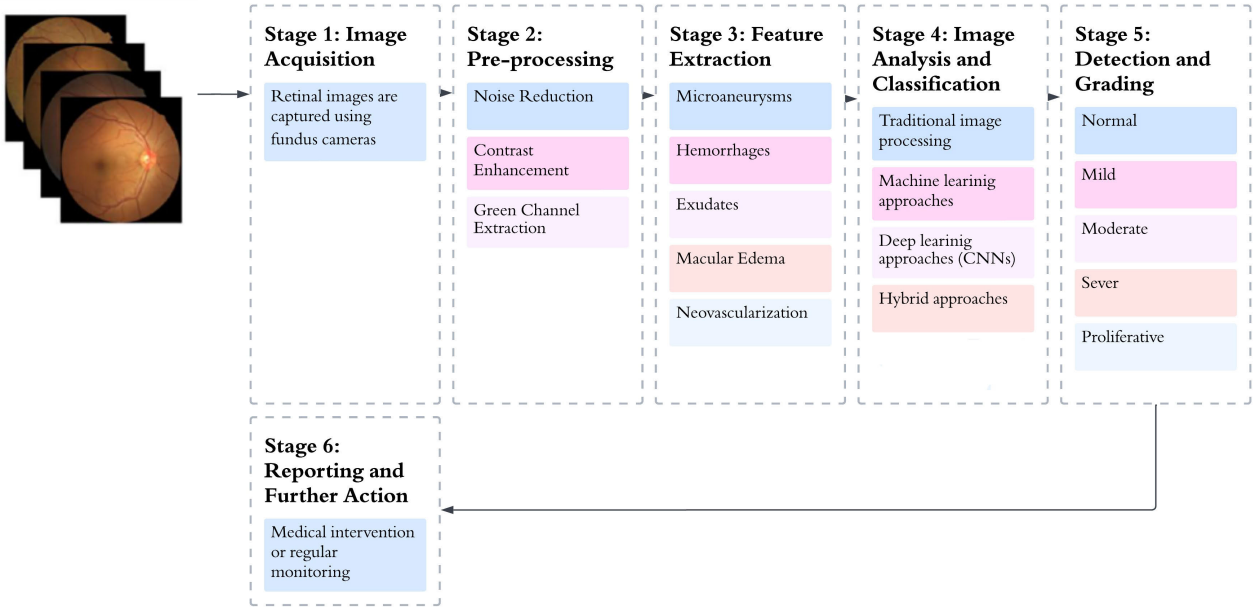


FIGURE 4. The evolution of machine learning and deep learning approaches in DR detection.

mainly focusing on new architectures that have been developed in the last three years. Figure 5 represents the general stages of DR detection and grading using ML and DL methods.

A. FUNDUS IMAGE PREPROCESSING

The preprocessing step for most CNN-based architectures comes in the form of image enhancement, salt and pepper noise removal, data augmentation, lesion segmentation, and region of interest (ROI) detection. Image enhancement consists of a few methods that can increase the contrast of the image such as contrast limited adaptive histogram equalization (CLAHE) [27], green channel extraction [28], and Gamma correction [29]. Noise removal is done using different types of filters such as median, mean, and Gaussian filtering [26]. For data augmentation, traditional rotation, sheering, mirroring, and zooming are used in most end-to-end learning methods such as those reported in [30], [31], [32], [33], and [34]. However, advanced DL-based methods such as generative adversarial networks (GAN) can also be used such as the one presented in [35]. For segmentation and ROI detection, architectures such as variants of U-Net [36] and regional convolutional neural networks [37] have been used in works [38] and [39]. These methods focus on identifying lesions in the fundus image which is later used in the classification of DR by learning from specific lesions and not the whole image, reducing the number of features to learn from. Archimedes optimization algorithm with Kapur's entropy has also been used for lesion segmentation in the work presented in [40].

B. EARLY DEEP LEARNING IN DIABETIC RETINOPATHY DETECTION

Some early contributions in DL for DR diagnosis were presented in [41], [42], [43], and [44]. Amongst these contributions, the work presented in [41] developed a deep learning algorithm to predict future DR progression using color fundus photographs. One of their models predicted at 12 months, achieved an AUC of 0.79, highlighting the potential of DL in predicting DR progression and aiding early diagnosis. The study conducted in [42] proposed a CNN approach for diagnosing DR, which demonstrated a sensitivity of 95% and an accuracy of 75% on the Kaggle dataset. This study emphasizes the benefits of automated systems in enhancing DR diagnosis efficiency. The work in [43] created a deep learning algorithm for automated DR detection, achieving a 0.97 AUC with 94% sensitivity and 98% specificity. This method represents a significant advancement in automated DR screening. A novel deep convolutional neural networks (DCNN) approach for DR diagnosis is presented in [44], which achieved an 86.17% recognition rate. Their method demonstrates the potential of DCNNs in early DR detection, particularly in resource-limited settings. The advantages and limitations of various approaches are summarized in Table 2.

C. TRANSFER AND ENSEMBLE LEARNING IN DIABETIC RETINOPATHY DETECTION AND GRADING

Various CNN architectures have been explored for DR detection, which are called specific instances of CNN architectures [45]. Often, medical image datasets are limited in size. Transfer learning addresses this by using pre-trained models on large datasets and fine-tuning them for specific tasks such as DR detection and grading. This approach allows models to leverage previously learned features, improving performance even with smaller datasets. The exploration of innovative approaches in diabetic retinopathy (DR) detection and classification has led to several significant studies, each employing unique methodologies and deep learning techniques to enhance accuracy and efficiency. These studies

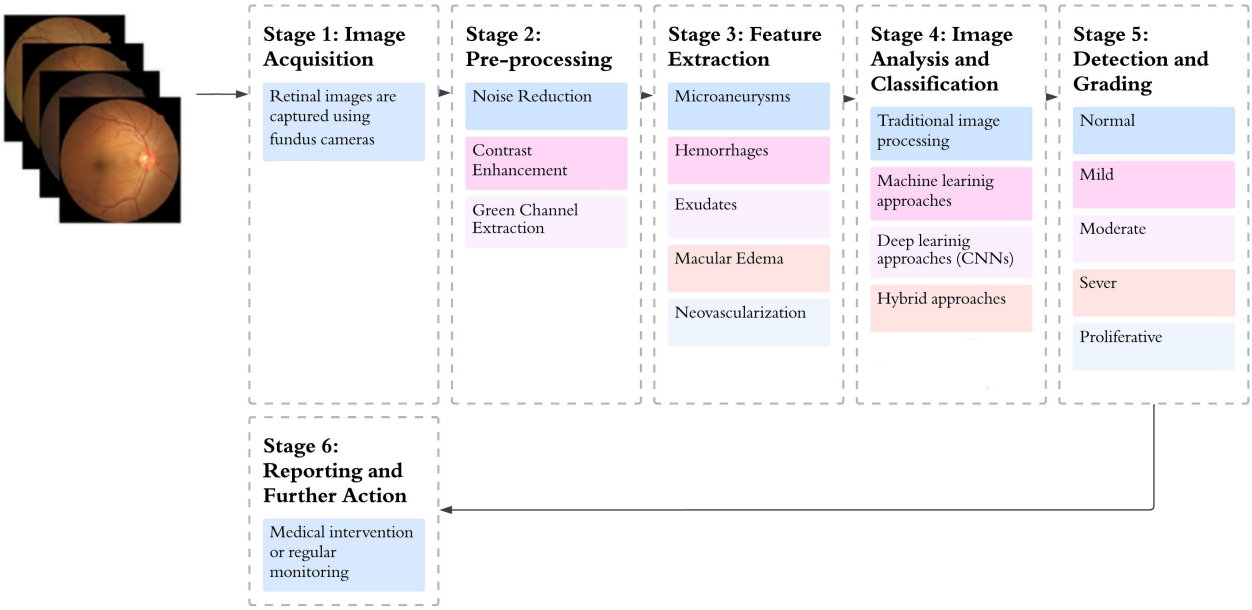


FIGURE 5. The general stages of detection and grading in diabetic retinopathy.

reflect the ongoing efforts to overcome the challenges in DR diagnosis and improve patient outcomes. Key contributions include the work in [46] which focused on the challenge of diverse retinal features across datasets in DR detection. The authors in [46] employed a transfer learning approach with four CNN models, among which the DenseNet121 [47] model excelled, achieving a 98.97% accuracy on a combined dataset. This study highlighted the benefits of combining datasets to enhance DR detection models. More studies such as the ones presented in [34], [39], [48], [49], [50], and [51] employ transfer learning of various sorts with pre-trained CNN modules such as AlexNet [52], DenseNet [47], ResNet [53], VGGNet [54], Inception-v3 [55], EfficientNet [56], and their variations. These methods usually retrain either the dense layers of the architecture or retrain the whole module on custom datasets for DR detection and grading. Apart from these networks, the Inception-ResNet [57], XceptionNet [58], NasNet [59], and MobileNet [60] can also be used as pre-trained models that are trained on ImageNet dataset [61]. Transfer learning yields good results for DR grading with some methods reaching over 90% over gold standard datasets such as APTOS [13], EyePacs [15], Messidor [14], and IDRiD [16]. In our work, we have provide extensive experimentation on transfer learning-based architectures to find the best model for DR detection and grading.

Ensemble learning has been introduced in DR diagnosis very recently with architectures such as the one presented in [38], [62], [63], and [64] being the most prominent ones in the last three years. The work in [64] presents the fusion of three pre-trained CNN architectures and couples it with a support vector machine to predict four different classes of DR. This method can be categorized within the transfer and ensemble learning paradigm as well as within the hybrid architecture genre. Monteiro proposed a method in his work presented in [62], namely, Blended deep learning that stacks ten different CNN architectures to predict the class of DR. Similarly, the work in [63], combined ResNet-34 [53], Inception-v3 [55], and VGG16 [54] and took a weighted mean of the three architectures for blood vessel segmentation leading to the detection of DR. The method presented in [38] used DenseNet-121 [47] in combination with ResNet-50 [53] to predict classes of DR after the segmentation of blood vessels using the improved version U-Net [36] architecture. Table 3 includes a few ensemble learning-based models in DR detection and grading with regard to their performances.

D. MODERN END-TO-END LEARNING CONVOLUTIONAL NEURAL NETWORK ARCHITECTURES IN DIABETIC RETINOPATHY CLASSIFICATION

Apart from transfer learning approaches, a few novel CNN-based methods for DR classification have been introduced in the field in recent times that employ end-to-end training on small datasets such as APTOS [13], EyePacs [15], Messidor [14], DDR [17], and IDRiD [16]. These architectures often employ augmentation to increase the data to train their proposed architectures. Examples of such architectures can be found in works presented in [30], [32], [33], [40], [72], [73], [74], [75], [76], [77], [78], and [79]. These architectures, though trained from scratch, show good learning logic and display high performance in the detection and grading of DR. Each of these architectures contains a novel factor integrated into traditional deep CNN methods to improve the architecture, or to design a new architecture that revolves around CNN encoders. Some examples of the improvement of traditional CNN-based methods can be found in [32], [33], [40], [72], [74], [75], and [77]. While the work in [72] uses the Inception [55] architecture as the backbone of its proposed method, the authors introduced long-range global

TABLE 2. Comparison between the DR detection and grading methods, highlighting the advantages and limitations of each method.

Models	Advantages	Limitations
Traditional Image Processing Techniques	✓Established and well-understood methods. ✓Can be effective for specific types of features or lesions ✓Do not require large, labeled datasets for training.	• Often require manual intervention or tuning. • May lack robustness for varying image qualities or sources. • Typically outperformed by deep learning methods in terms of accuracy.
Machine Learning Approaches	✓Offer flexibility in choosing features and classifiers. ✓Can be trained in moderate-sized datasets. ✓Transparent and interpretable decision-making process.	• Require manual feature extraction and selection. • Performance can be sensitive to feature engineering quality. • Typically require more preprocessing steps than deep learning.
Deep Learning	✓State-of-the-art performance in many DR detection tasks. ✓Can learn features directly from raw data, eliminating manual feature engineering. ✓Scalable with data: performance often improves with more data.	• Often acts as a "black box" with limited interpretability. • Can be computationally intensive, especially for complex architectures.
Hybrid Approaches	✓Combine the strengths of both traditional and deep learning methods. ✓Allow for innovative combinations and methodologies. ✓Can lead to improved robustness and accuracy.	• Can be more complex to implement and fine-tune. • Might inherit challenges from both traditional and deep learning methods.

dependence units to connect feature maps from different layers of convolution. This improves the model significantly by enhancing the learning on small datasets and avoids overfitting. Similarly, the work in [74] treats the multiclass classification task as two binary and one ternary classification task. Three networks have been used to achieve DR grading by dividing the detection tasks that classify images as DR and No-DR, NPDR and PDR, and then finally as mild, moderate, and severe NPDR. Each network utilized a separate architecture with its separate preprocessing steps allowing the model to rectify the class imbalance contained in the datasets. The work in [40] introduces the chimp optimization algorithm to train a variant of DenseNet [47] with a high number of layers. A fresh CNN module, namely, Spiking neural networks for multiclass classification tasks of DR grading was also introduced in the paper. Other methods involving residual block integration, entropy enhancement, and discrete wavelet transform have been coupled with traditional CNNs in works [16], [32], [75], [77] enabling them to be trained from scratch rather than using pre-trained models.

Some unconventional CNN-based DL architectures were presented in [30], [73], [76], [78], and [79]. These architectures do not follow the traditional way of stacking pooled layers. Rather, those works introduced newer and more advanced ways of cascading skip connections, extracted feature maps, and attention layers while improving the activation and loss functions. The work in [30], introduces the integration of skip connections in dilated convolutional blocks and uses hierarchical feature extraction to generate feature maps from these convolutional layers. The work in [73] presented a novel method that improves existing capsule networks [80] and suits them to DR detection and grading tasks. The works presented in [78] and [79] present newer and shorter versions of the ResNet [53] that are suitable for learning from low data-containing datasets. These architectures show better generalization capability and lower fluctuations on the learning curve compared to traditional ResNets with 50 layers or more. Table 3 shows the comprehensive survey of some end-to-end learning CNN architectures highlighting their performances in DR detection and classification.

E. HYBRID AND ADVANCED MODELS

Hybrid models combine CNNs with other algorithms such as extreme learning machines (ELM), transformers, traditional ML methods such as support vector machine (SVM), or boosting methods to produce a faster and more reliable learning architecture. Some of these methods can be found in the works presented in [28], [31], [64], [65], [66], [67], [68], [69], [70], and [71]. The study published in [67] explored the combination of VGG16 and XGBoost architectures for DR detection using the APTOS [13] dataset. The work in [66] introduced an Extreme Learning Machine (ELM) approach and coupled it with CNN and singular value decomposition-based feature selection for DR diagnosis, utilizing the APTOS [13] and Messidor [14] datasets. Their method achieved impressive accuracies of 99.73% for binary classification and 98.09% and 96.26% for multiclass classification on these respective datasets, demonstrating the effectiveness of ELM in DR diagnosis. The work in [66]

TABLE 3. Comprehensive performance analysis of state-of-the-art DL methods in DR grading.

Genre	Reference	Model	Dataset	Grading Accuracy (%)
Transfer Learning	Mutawa *et al.* [46]	DenseNet -121	APTOS + EyePACS	98.97
	Das *et al.* [48]	EfficientNetB4	EyePACS	79.11
	Ohri *et al.* [49]	ResNet-50	APTOS	80.10
	Erciyas *et al.* [39]	VGG19	EyePACS / Messidor	100 / 100
	Lin *et al.* [50]	ResNet-50	APTOS / EyePACS	-
	Zhang *et al.* [51]	Inception-v3	EyePACS	-
	Patil *et al.* [34]	ResNet-50	APTOS + EyePACS	97.87
Ensemble Learning	Monteiro [62]	10 pre-trained CNN Modules	DDR + EyePACS + Messidor2 + APTOS	-
	Mary *et al.* [38]	Densenet-121 + Resnet-50	APTOS	98.69
Hybrid Learning	Jabbar *et al.* [28]	GoogleNet + ResNet	EyePACS	94.00
	Parthiban *et al.* [65]	SqueezeNet	Messidor	99.65
	Nahiduzzaman *et al.* [66]	CNN + SVD + ELM	Messidor2 / APTOS	98.09 / 96.26
	Mohanty *et al.* [67]	VGG16 + XGBoost	APTOS	79.50
	Dai *et al.* [31]	SWIN transformer	EyePACS	87.43
	Bhimavarapu *et al.* [68]	ResNet-50 + SVM	APTOS / EyePACS	99.56 / 99.97
	Nahiduzzaman *et al.* [69]	PCNN +ELM	APTOS / EyePACS	97.27 / 91.78
	Ma *et al.* [70]	Transformer + CNN	IDRiD	77.82
	Bala *et al.* [71]	Transformer + CNN + Res	APTOS / IDRiD	98.60 / 98.90
Ensemble + Hybrid Learning	Thomas *et al.* [64]	Triple CNN + SVM	EyePACS + Messidor + Private	98.94
End-to-End Learning	Luo *et al.* [72]	CNN + local / long range dependence	EyePACS / Messidor	83.60
	Khan *et al.* [30]	Ddnet	APTOS	97.00
	Kalyani *et al.* [73]	Capsule Networks	Messidor	-
	Ashwini *et al.* [32]	DWT + CNN	DDR / EyePACS / IDRiD	96.20 / 93.5 / 90.07
	Jian *et al.* [74]	Triple-DRNet	APTOS	92.08
	Ragab *et al.* [40]	DenseNet-264 + SpikinNet	Messidor	99.73
	Li *et al.* [75]	Lesion-attention pyramid	Messidor / EyePACS	92.30 / 89.10
	Yue *et al.* [76]	Attention-driven Cascaded Network	APTOS / EyePACS	83.4 / 72.44
	Imran *et al.* [77]	DWT + Dual CNN + Residual Blocks	Messidor2 / APTOS	80.00 / 92.00
	Sunkari *et al.* [78]	ResNet18 + Swish activation function	APTOS	95.50
	Raiaan *et al.* [79]	ResNet10	APTOS + IDRiD + Messidor	98.65
	Kommaraju *et al.* [33]	CNN + Residual Blocks	DRR	96.23

was further enhanced in [69] by proposing a parallel dual CNN-ELM architecture. This method can get rid of the computationally complex singular value decomposition and produce better results in DR classification. The work presented in [65] proposed a deep ELM framework. However, the learning curve obtained from this method is not stable and therefore the use of deep ELM in DR classification can be regarded as questionable. Several methods such as the ones presented in [28], [64], and [68] have combined CNN with SVM to form a hybrid architecture for the prediction of DR. These methods usually use the CNN module to extract important structural features while they use SVM to achieve classification.

Transformers have become the new state-of-the-art modules in DL-based classification tasks in recent years. Methods presented in [31], [70], and [71] utilize the power of the attention modules present in transformers to predict stages of DR. Transformers are new architectures that combine CNNs with attention to acquire better feature maps than traditional CNNs. However, because the use of transformers in medical

image classification is a very raw stage and there are very few pre-trained transformer models, most methods fall short of good performance. The highest accuracy achieved with transformers has been reported in [71] which is 98.6% and 98.9%, respectively, for the APTOS [13] and IDRiD [16] datasets. This method combines a CNN module with residual connection (Res) to achieve such performance indicating that transformers when used alone may result in lower performance compared to their CNN counterparts. This is further proven by the works in [31] and [70], where the classification accuracy suffers tremendously when a transformer is utilized without a CNN module to support it. Table 3 shows the performance of all hybrid and advanced models that have been surveyed in this work.

IV. COMPARISON METHODOLOGY

This section describes the methodology adapted to the testing of various CNN architectures by applying transfer learning to them. From the previous sections, it can be seen that even though highly sophisticated methods are being developed, the use of various pre-trained DL architectures has not been exclusively experimented upon. Furthermore, as seen from the previously surveyed literature, most methods either combine several datasets into one grand one for training and testing or train and test their architectures individually on each dataset. This raises the question of whether a neural network being trained on one dataset can perform well on another. Therefore, in this work, we experimentally verify the effectiveness of several pre-trained architectures by retraining them on one dataset and then using the best model to perform tests on a second dataset. The notion is to find out if a pre-trained model that is retrained on a low amount of DR data can perform well on a completely separate dataset.

A. DATASETS IDENTIFICATION

For experimentation, modified versions of two relatively large datasets are taken, namely, the APTOS [13] and the EyePACS [15] datasets. The data is then preprocessed and augmentation is utilized in the training phase.

1) DIABETIC RETINOPATHY GAUSSIAN FILTERED (DRGF)

This dataset consists of 3,662 Gaussian-filtered retina scan images. There are a total of five categories of fundus images included in this dataset according to the severity of diabetic retinopathy. The dataset is the resized version of the APTOS 2019 Blindness Detection [13] and harbors images in the size of 224 $\times$ 224 pixels.

2) DIABETIC RETINOPATHY RESIZED (DRR)

This dataset contains all instances from the training set of the EyePACS [15] dataset. Essentially the EyePACS dataset is available in two subsets, train and test, containing 32,126 and 53,576 images, respectively. For our experimentation purposes, we have only taken the training set which is available publicly in the form of the Diabetic Retinopathy Resized (DRR) dataset in Kaggle. The images in this dataset are also categorized into five different categories according to the severity of diabetic retinopathy.

B. DATA PREPROCESSING

The 'createGen' function is used; it defines the data generator for training, validation, and testing. These generators preprocess the images before feeding them into the model. The training generator also includes data augmentation using rotation, zooming, shearing, width and height shift, and horizontal flipping. Rotation ranges of -30 and 30 degrees are used to produce rotated copies of the original fundus image. A zoom of 0.15 has also been utilized to keep most of the information in the image intact. Horizontal and vertical shifts to the fundus image have also been utilized in the range of -20 pixels to 20 pixels. A shearing range of 0.15 has been utilized to make modified copies of the fundus image. Finally, a horizontally flipped copy of the image is made to enhance the dataset.

After these data augmentation techniques are applied to the images to generate new transformed images that help the model to generalize better, the dataset is split into training and validation sets. The data is shuffled and stratified to maintain a balanced distribution of the classes. The 'createGen' function also creates iterators for training and validation sets using the 'flowfromdataframe' method. This method reads images from the file paths specified in the Data Frame, applies the necessary preprocessing and augmentation, and yields batches of images and labels to be used in training and validation.

C. TRANSFER LEARNING ON PRE-TRAINED MODELS

This study implemented 26 pre-trained models from the TensorFlow Keras applications module, namely (DenseNet121, DenseNet169, DenseNet201 [47], EfficientNetB0, EfficientNetB1, EfficientNetB2, EfficientNetB3, EfficientNetB4, EfficientNetB5, EfficientNetB6, EfficientNetB7 [56], InceptionResNetV2 [57], InceptionV3 [55], MobileNet, MobileNetV2, MobileNetV3Large, MobileNetV3Small [60], NASNetMobile [59], ResNet101, ResNet101V2, ResNet152, ResNet152V2, ResNet50 [53], VGG16, VGG19 [54], and Xception [58]). The structure of the implemented models is represented in Figure 6. These models have been previously trained on the ImageNet dataset [61] and serve as effective feature extractors for DR grading and detection tasks. Figure 7 shows an example of transfer learning in DR diagnosis. Each pre-trained model is modified by adding two fully connected layers with 128 units and ReLU activation, followed by a softmax output layer for multi-class classification. The pre-trained models' layers are set to non-trainable, ensuring that only the newly added layers are updated during training. Each model is trained twice to test the performance once on the binary classification of DR and a second time on the multiclass classification of DR stages. In the first step, the model is trained to perform DR detection. In this step, only two labels are available, namely, DR and No DR. In the second step, the model is trained to perform multiclass classification to identify the stages of DR. All training is done using the first

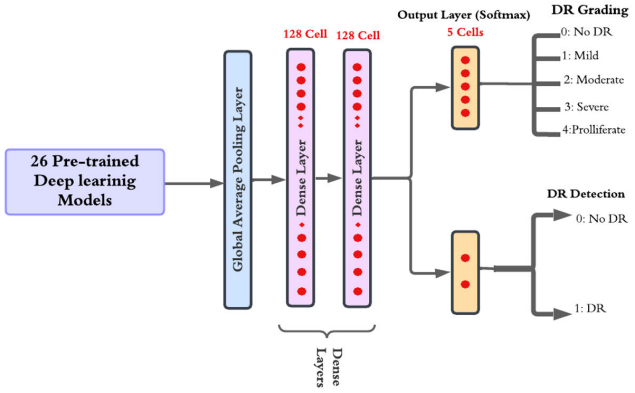


FIGURE 6. The general stages to detect and grade DR.

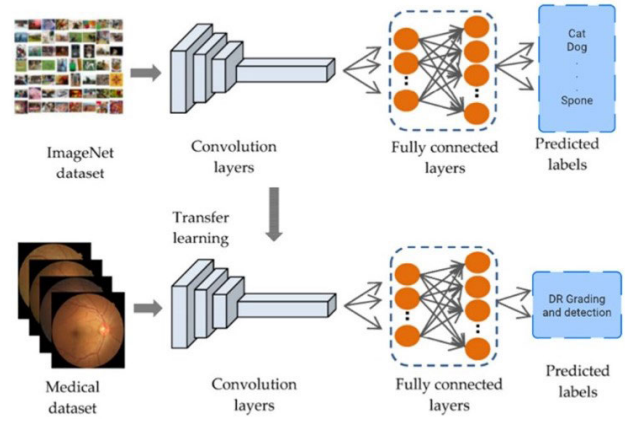


FIGURE 7. Transfer learning in DR diagnosis.

dataset (DRGF), and after choosing the best model, the tests employ the second dataset (DRR).

For each model to run, the images are reshaped according to the input size criteria of the models. Batch normalization is used before passing the images through the convolutional layers. For binary classification, the binary-cross entropy loss (BCE) is employed, while for grading or multiclass classification categorical cross entropy (CE) is used. The learning rate is set to 0.00001 for each model and a batch size of 256 is employed. Each modified model is trained for 10 epochs using the training dataset and validates its performance on the validation dataset. The performance of each of the 26 models was compared using the DRGF dataset in terms of training accuracy, validation accuracy, and training time.

Once the models are trained, Gradient-weighted Class Activation Mapping (Grad-CAM) [81] is used to view the heat maps generated for the classification performed on the validation set. Grad-CAM is a visualization technique used in deep learning to understand and interpret the decision-making process of CNNs. It helps in identifying the important regions in the input image that the model focuses on when making its predictions. This technique has been widely used for tasks such as image classification, object detection, and segmentation. Grad-CAM works by computing the gradient

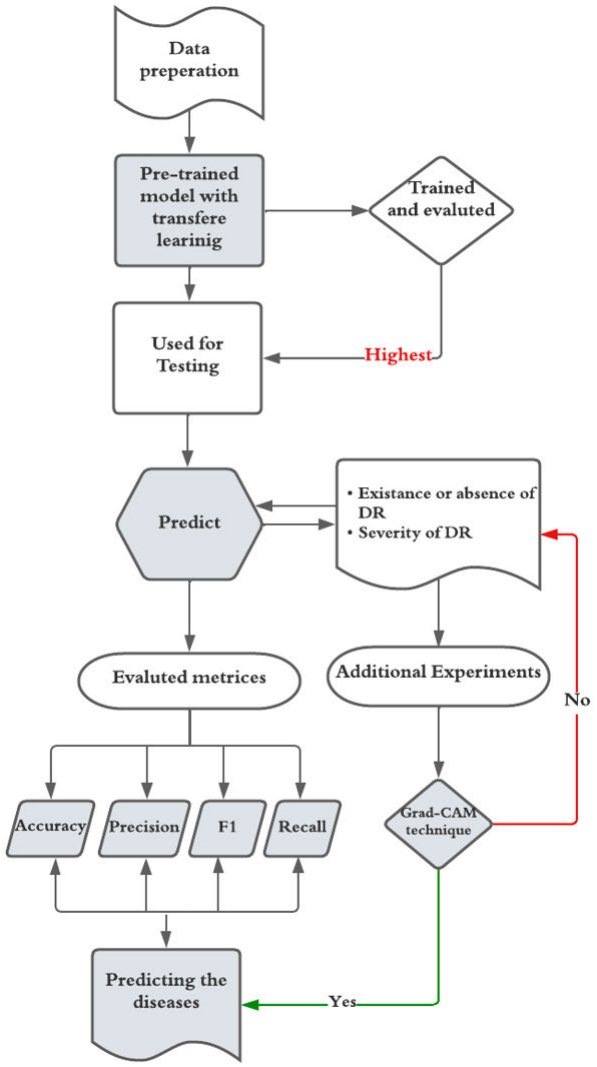


FIGURE 8. Block diagram of the proposed comparison methodology.

of the predicted class score with respect to the feature maps of the last convolutional layer in the CNN. These gradients are then pooled to create weights that signify the importance of each feature map for a particular class prediction. The weighted combination of the feature maps, followed by a ReLU activation, generates the final Grad-CAM heatmap. By generating heatmaps, Grad-CAM helps physicians to understand which parts of the retina scan images the model finds most relevant for making predictions of the DR severity. The methodology framework of the proposed DR detection and grading system is illustrated in Figure 8 and examples of Grad-CAM visualizations of heat maps are provided in Figure 9. These heat maps have been produced from the validation set of the DRGF dataset.

V. EXPERIMENTAL RESULTS AND DISCUSSION

In this section, the results from the proposed comparison methodology are presented followed by an analysis of the results obtained. The performance comparison of different

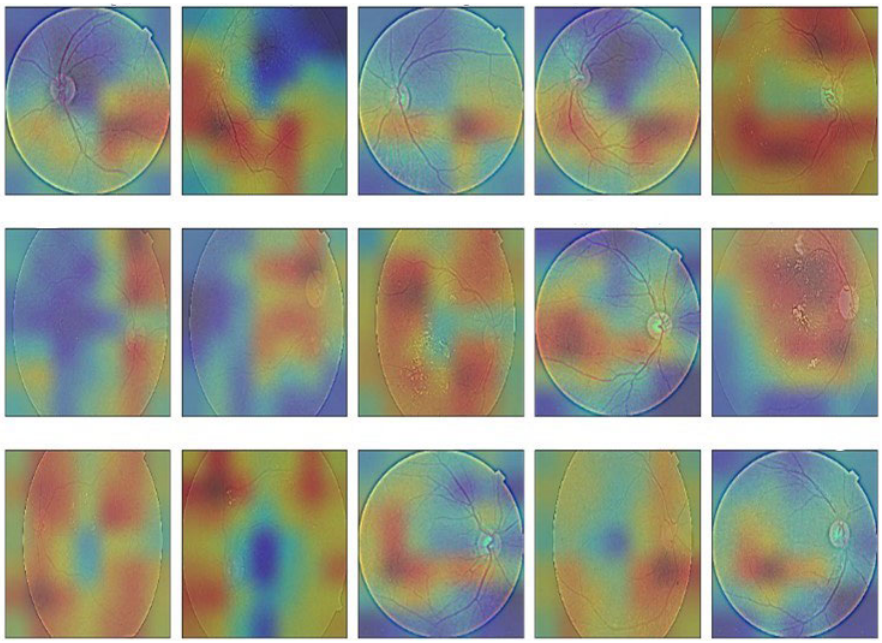


FIGURE 9. Grad-CAM visualization of fundus image data from the validation set of the DRGF dataset.

pre-trained models with respect to their training accuracy, test accuracy, and training time is also detailed in this section.

A. EVALUATION METRICS AND IMPLEMENTATION DETAILS

The effectiveness of models in DR detection and classification is assessed using various metrics, each offering a different perspective on the model's performance. These metrics are accuracy (*Acc*), recall (*Rec*), precision (*Prec*), and harmonic mean of precision and recall ($F1$) scores. The accuracy is defined as the total correctly classified events divided by the total number of classified events, while the recall score is the fraction of the number of true positives divided by the total actual positives, the precision score is the fraction of the actual correct prediction divided by the total predictions, and the $F1$ score is the weighted average of both the recall and precision scores. These metrics are represented by the following equations,

$$Acc = \frac{TP + TN}{TP + TN + FP + FN}, \tag{1}$$

$$Rec = \frac{TP}{TP + FN}, \tag{2}$$

$$Prec = \frac{TP}{TP + FP}, \tag{3}$$

$$F1 = \frac{2 \times TP}{2 \times TP + FP + FN}, \tag{4}$$

where *TP* and *TN* represent the True positive and true negative classification cases, respectively, while *FP* and *FN* represent the False positive and false negative cases, respectively.

All experiments have been conducted in Python using Jupyter Notebook. The operating system used is Windows 11, 11th Gen Intel(R) Core(TM) i7 with random access memory (RAM) of 12 GB. Both Keras and TensorFlow have been utilized to train all models and Grad-CAM implementation. All pre-trained models have been imported from the module 'tensorflow.keras.models' and the training of the convolutional layers has been set to false.

B. VALIDATING THE PERFORMANCE OF 26 PRE-TRAINED MODELS

The performance of each of the 26 models is compared using the DRGF dataset in terms of training accuracy, validation accuracy, and training time per epoch on multiclass classification. The DRGF database was split into training and validation sets with 90% for training and 10% for validation, respectively. Table 4 represents the maximum performance in accuracy obtained by each of the 26 pre-trained models.

It is seen that some architectures and their variants such as DenseNet, Xception, NasNetMobile, and Inception show high accuracy in both training a validation with only 10 epochs of training with DenseNet201 reaching the highest accuracy of 81.76% on the validation set from the DRGF dataset. However, this architecture is very complex with 201 total layers which require approximately 25% more time to train compared to MobileNet which requires the least amount of training time. The highest training accuracy is obtained by ResNet152V2. Even though ResNet152V2 exhibits a stronger performance in training, the validation accuracy displayed by the architecture is approximately 11 % less than that achieved by DenseNet201. The highest training times are exhibited by DenseNet121, ResNet152V2, and ResNet152 with InceptionResNetV2 being right at their heels. On the other hand, models from the EfficientNet family tree exhibited lower performance, with validation accuracies ranging from 30.70% to 56.53% while the time for training these architectures is moderate with an average of approximately 14.5 seconds per epoch. The

TABLE 4. The comparison of performance of various pre-trained models on the training dataset (DRGF).

Model	Training Accuracy (%)	Validation Accuracy (%)	Training time (sec)
DenseNet121	81.66	78.12	201.99
DenseNet169	84.22	77.81	141.75
DenseNet201	83.14	81.76	155.67
MobileNet	84.05	76.90	125.41
MobileNetV2	85.57	75.99	130.01
MobileNetV3Small	67.26	69.91	127.72
MobileNetV3Large	67.46	66.57	129.28
Xception	83.78	75.68	132.33
NASNetMobile	80.55	79.03	152.61
VGG16	76.16	75.08	141.42
VGG19	75.72	72.64	133.12
InceptionV3	84.83	72.34	151.78
InceptionResNetV2	78.83	72.34	161.17
ResNet50	71.48	72.04	142.94
ResNet101	70.73	72.34	142.29
ResNet101V2	86.58	75.38	158.84
ResNet152V2	88.17	71.12	170.81
ResNet152	70.30	70.52	172.87
EfficientNetB3	54.45	56.53	147.83
EfficientNetB0	49.36	47.72	129.22
EfficientNetB4	49.36	47.72	146.29
EfficientNetB5	49.36	47.72	147.25
EfficientNetB2	49.36	47.72	134.08
EfficientNetB7	49.36	47.72	151.71
EfficientNetB6	49.36	47.72	144.77
EfficientNetB1	49.56	30.70	131.75

ResNet family displays mixed results in terms of training and validation accuracy. Large and small versions of MobileNet do not perform well on low data while all members of the DenseNet family show good performance on both training and validation sets. The Xception and NASNetMobile display a training set accuracy of over 80%. However, the difference in training and validation performances for the Xception module is approximately 8% while the difference is close to 1% for NASNetMobile.In terms of training time, Xception is approximately 15% faster than NASNetMobile due to its lower number of trainable parameters in the dense layer compared to NASNetMobile. Furthermore, MobileNetV2 is found to be the most computationally efficient among all architectures given that it shows the best average performance. Its architecture is based on depthwise separable convolutions that enable it to capture relevant features in images with fewer parameters and computations compared to other architectures. Therefore, this module is selected for testing on the DRR dataset for both DR detection and grading.

C. DR DETECTION AND GRADING USING DRR DATASET FOR THE MOBILENETV2 MODEL

Figure 10 presents various results for DR detection and grading out of experimentation while Figure 11 presents the Grad-CAM visualization of different types of DR from the DRR dataset. For the DRR dataset, the model achieved an average accuracy of 75% in the grading task. The model's performance was the highest for the detection task, with an average accuracy of 78%, precision of 77%, recall of 78%, and an F1-score of 73%. Table 5 introduces the performance evaluation of the pre-trained MobileNetV2 model for different classes from the DRR dataset. From Table 5, it can be seen that when the model is deployed for the multiclass grading task, it struggles to perform on unseen data. A precision of 77%, recall of 98%, and an $F1$-score of 86% were obtained for the No_DR class for grading. However, the performance on the classification of any type of DR is low for the pre-trained model. The model also showed good performance in the binary classification of DR. In the detection of the "No_DR" class, an $F1$-score of 86% was achieved, but its performance in detecting the "DR" class was lower, with an $F1$-score of only 36%. This might indicate that the model has some difficulty in detecting the presence of diabetic retinopathy in the DRR dataset not seen by the model in its training stage. One important note is that the training dataset, DRGF, has severe class imbalance which results in the model struggling to learn from the underrepresented classes, leading to a decrease in overall performance when deployed to another dataset such as the DRR. Also, from the experimentation, it can be seen that the Gaussian filtering process applied to the DRGF dataset at

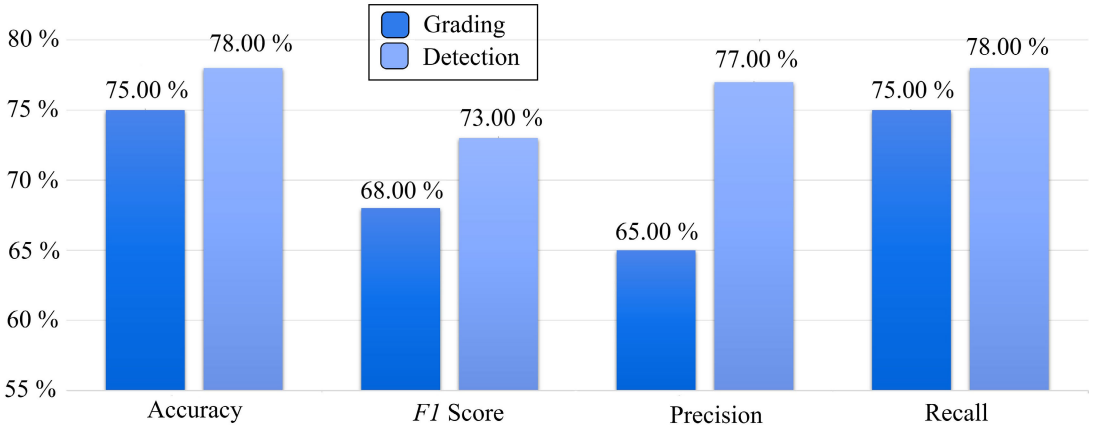


FIGURE 10. Performance of MobileNetV2 using the DRR dataset.

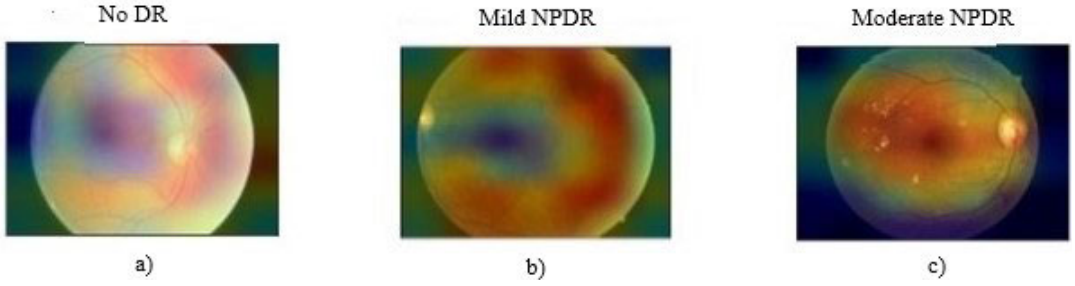


FIGURE 11. Grad-CAM visualization of samples of different types of DR from the DRR dataset.

TABLE 5. Performance evaluation of pre-trained MobileNetV2 model for difference classes using the DRR dataset.

DRR Dataset	Class	*Prec*	*Rec*	*F1*
Grading	Moderate_NPDR	41%	10%	16%
	No_DR	77%	98%	86%
	Proliferate_DR	51%	51%	51%
	Severe_NPDR	54%	15%	23%
	Avg/Total	**65%**	**75%**	**68%**
Detection	DR	73%	24%	36%
	No_DR	78%	97%	86%
	Avg/Total	**77%**	**78%**	**73%**

the preprocessing step has helped highlight important features in the images, making it easier for the model to learn and detect diabetic retinopathy. On the other hand, the dataset, DRR has not undergone any type of filtration process and therefore contains more data variability and noise, making it more challenging for the model to perform well on it.

Grad-CAM is used to visualize the regions of the input image that contribute the most to the classification. It does this by generating heat maps that highlight the important areas in the images. Figure 11 demonstrates the Grad-CAM visualization of the predicted retinal scan images from the DRR dataset for the grading task. Grad-CAM is a powerful visualization technique that allows interpretation of the decision-making process of the MobileNetV2 model for DR detection and grading. By examining the color scale, overlay, intensity, and spatial distribution of important regions in the heat map, we can gain insights into the model's focus areas and the factors influencing its classification decisions. The higher the intensity (closer to red), the more important that region is for the model's classification decision. It is seen from the Grad-CAM visualization that the trained model can capture certain structures such as exudates and blood vessels. Due to the imbalance of data in the dataset, the model captures with more efficiency the classes with more data. As seen from Figure 11 (a) and 11 (c), the model emphasizes only the important regions to make a decision. However, for classes with lower data such as Mild NPDR, the model looks at most of the important regions of the image while not able to make a rigid decision on which class it belongs to. This notion can be verified from Figure 11 (b) where the model searches for important features throughout the image.

VI. CONCLUSION

This survey has comprehensively explored the evolving landscape of machine learning and deep learning in the detection and classification of DR, a critical public health issue linked to the global rise of diabetes. The key findings from this survey highlight the significant progress and the potential

of these technologies in revolutionizing the approach to DR screening and diagnosis. Furthermore, several state-of-the-art methods that use DL as the backbone of their architecture have been surveyed. Additionally, the potential of the pre-trained models has also been explored for automated detection and grading of DR. Cross-dataset evaluation has been performed to verify the robustness of these architectures and if they can be used in DR diagnosis models. The Grad-CAM visualization technique has been used to get an insight into the model's decision-making process, helping to understand its focus areas and the rationale behind the classification decisions. This interpretability is crucial for building trust in the model's predictions and ensuring that practitioners can confidently rely on its outputs. Consequently, from the survey and experimentation, it can be concluded that the transition from traditional ML techniques to more advanced DL models, particularly CNNs, has marked a significant advancement in DR detection. This shift has enabled more accurate, efficient, and automated analysis of retinal images. The exploration of various CNN architectures, such as VGG, ResNet, and Inception models, has demonstrated their effectiveness in capturing complex patterns in retinal images for accurate DR detection through transfer learning. To improve the diagnosis even further, data augmentation strategies and models are also being built to enhance the performance and generalizability of new end-to-end learnable DL models. The development of hybrid models, which combine CNNs with other techniques such as SVD, SVM, and transformers, has also shown effectiveness in improving feature extraction and classification accuracy. However, while model generalization capability is high when tested on a portion of the same dataset, the model usually suffers when deployed on a different dataset. More future research is required for data preprocessing that would remove unwanted bias from datasets. Future research may explore the integration of additional data sources, such as patient demographics or medical history, to enhance the model's performance.

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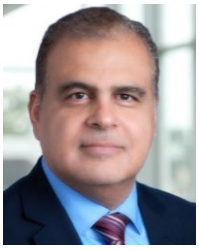
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